COLLIGATIVE PROPERTIES: FREEZING POINT DEPRESSION

OBJECTIVES

- To explore the colligative properties of freezing point depression
- To determine the effects of solution molality on colligative properties
- To compare the effects of electrolytes and nonelectrolytes on colligative properties

INTRODUCTION

The world we live in is filled with solutions. Oceans, lakes, rivers and streams and the fluids that move in the bodies of living organisms are all solutions. The air we breathe is a solution of gases and the automobiles, trains, planes and bicycles we use for travel are solutions of solids. Brass door handles are solutions of zinc in copper. Thus, most of the materials we come in contact with during our daily experiences are mixtures. Homogeneous mixtures (mixtures with uniformly mixed components) are called solutions. More specifically, homogeneous mixtures contain two or more substances. The major component in a solution is called the solvent, (e.g. water is the solvent in our oceans) and the minor component(s) are called the solute(s) (e.g. sodium ions and chloride ions in seawater). Because the primary component of solutions is solvent, the physical properties of the solution resemble those of the pure solvent. However, some physical properties, called colligative properties, are independent of the nature of the solute and depend only upon the concentration of solute particles. Examples of colligative properties include vapor pressure reduction, boiling point elevation, freezing point depression, and osmotic pressure.

You are probably familiar with some common examples of these effects. Pure water freezes at 0.0℃ (32°F) and boils at 100℃ (212°F) when the atmospheric pressure is 1 atm. But, in the presence of a solute such as ethylene glycol (antifreeze) the freezing and boiling point values are altered. Salt also affects the melting point of water, which is why it is used to melt ice on roadways, driveways and sidewalks.

These effects are expressed quantitatively by the *colligative-property law*, which states that the freezing point and boiling point of a solution differ from those of the pure solvent by amounts that are directly proportional to the *molal* concentration of the solute (molal concentration or molality, *m*, is expressed in units of moles solute per kg solvent). In its most *general* form, this equation is written:

$$\Delta T = K \times m \qquad (1)$$

where ΔT is the boiling point elevation or freezing point depression, K is a constant that is specific for each solvent, and m is the molality of the solution.

Table 1: K_{fp} freezing point and K_{bp} boiling point for a variety of solvents.

Solvent	**Chemical formula**	**Freezing Point** (°C)	$\boldsymbol{K_{fp}\left(\frac{°C \times kg}{mol}\right)}$	**Boiling Point** (°C)	$\boldsymbol{K_{bp}\left(\frac{°C \times kg}{mol}\right)}$
Acetic acid	CH_3CO_2H	16.6	3.90	118.1	2.93
Benzene	C_6H_6	5.4	5.12	80.2	2.53
Chloroform	$CHCl_3$	−63.5	4.68	61.3	3.63
Ethanol	C_2H_5OH	−114.6	1.99	78.4	1.22
Water	H_2O	0.0	1.86	100.0	0.512
Naphthalene	$C_{10}H_8$	80.6	6.90	218.0	5.65
Cyclohexane	C_6H_{12}	6.5	20.0	81.0	2.79
Camphor	$C_{10}H_{16}O$	179	39.7	208.0	5.95

Boiling point elevation and freezing point depression can be related to molality by:

$$\Delta T_{bp} = K_{bp} \times m \times i \quad (2)$$

$$\Delta T_{fp} = K_{fp} \times m \times i \quad (3)$$

Where ΔT_{bp} and ΔT_{fp} are the change in boiling point and freezing point respectively, K_{bp} and K_{fp} are the molal boiling-point-elevation and molal freezing-point-depression constants for a particular solvent, and ***i* is the van't Hoff factor that is used to account for the number of particles a given solute generates in solution**.

Substances are usually classified as ionic or covalent—or more precisely, *electrolytes and non-electrolytes*. Electrolytes dissolve in a solvent to give IONS (charged particles) in solution while non-electrolytes dissolve to give MOLECULES. A mole of non-electrolyte such as ethanol would dissolve to produce a mole of ethanol molecules. However, a mole of sodium chloride ($NaCl$) would dissociate to form two moles of ions ($Na^+$and Cl^-). Therefore, a 1 m aqueous solution of $NaCl$ is 1 m in Na^+ and 1 m in Cl^-, making it 2 m in total solute particles. Because colligative properties are related to the number of solute particles, we would expect a 1 molal solution of NaCl to have twice the effect as a 1 molal solution of ethanol. Therefore, $i = 1$ for ethanol and other nonelectrolytes, $i = 2$ for $NaCl$, and $i = 5$ for $Al_2(CO_3)_3$.

As an example, let's say that we dissolve 10 g of the solute ethanol (C_2H_5OH) in 100 g of pure water (the solvent). What would be the boiling point, and freezing point of this solution? First, we must determine the molality (m) of the solution. To do this we need to know the moles of solute and kilograms (kg) of solvent:

$$10\ g\ C_2H_5OH \times \frac{1\ mol\ C_2H_5OH}{46.069\ g} = 0.217\ mol\ C_2H_5OH$$

$$molality\ of\ C_2H_5OH\ \left(m_{C_2H_5OH}\right) = \frac{mol\ C_2H_5OH}{kg\ H_2O} = \frac{0.217\ mol\ C_2H_5OH}{0.100\ kg} = 2.17\ m$$

For water the $K_{bp} = 0.512\ \frac{℃}{m}$, and $K_{fp} = 1.86\ \frac{℃}{m}$. So, for this ethanol solution:

$$\Delta T_{bp} = \left(0.512\ \frac{℃}{m}\right) \times (2.17\ m) \times (1) = 1.11\ ℃$$

the boiling point is increased by $1.11℃$ to $101.11℃$.

Likewise, the freezing point:

$$\Delta T_{fp} = \left(1.86\ \frac{℃}{m}\right) \times (2.17\ m) \times (1) = 4.04\ ℃$$

is decreased by $4.04℃$ to $-4.04℃$. **It is important to not confuse the actual freezing point (T_{fp}) with the number of degrees that the freezing point is depressed (ΔT_{fp})!**

In the above example, we assumed that the van't Hoff factor for ethanol had a value of one ($i = 1$). Let us assume that the same ethanol/water solution had an experimentally determined freezing point of $-3.75℃$. We could calculate the *actual* van't Hoff factor as follows:

$$\Delta T_{fp} = K_{fp} \times m \times i$$

$$\Delta T_{fp} = T_{solvent} - T_{solution}$$

Rearrange to solve for i:

$$i = \frac{\Delta T_{fp}}{K_{fp} \times m}$$

$$i = \frac{3.75\ ℃}{(1.86\ \frac{℃}{m}) \times (2.17\ m)} = 0.93$$

In this lab, you will experimentally determine the van't Hoff (i) factor for two different substances, sucrose ($C_{12}H_{22}O_{11}$) and sodium chloride ($NaCl$). From the discussion above you should be able to ascertain these values. However, these theoretical values are only valid for very dilute solutions (<0.01 m). At higher concentrations, the ions do not behave independently and so the values of (i) will be ***less*** than theoretical.

In this experiment you will need to determine the freezing point depression for two solutions. However, the temperature at which a solution freezes is difficult to determine by direct visual observation because of a phenomenon called supercooling and also because solidification of solutions usually occurs over a broad temperature range. To overcome these problems, temperature-time graphs, called *cooling curves*, are used which readily reveal the true freezing temperature.

The major features of a general cooling curves are shown below:

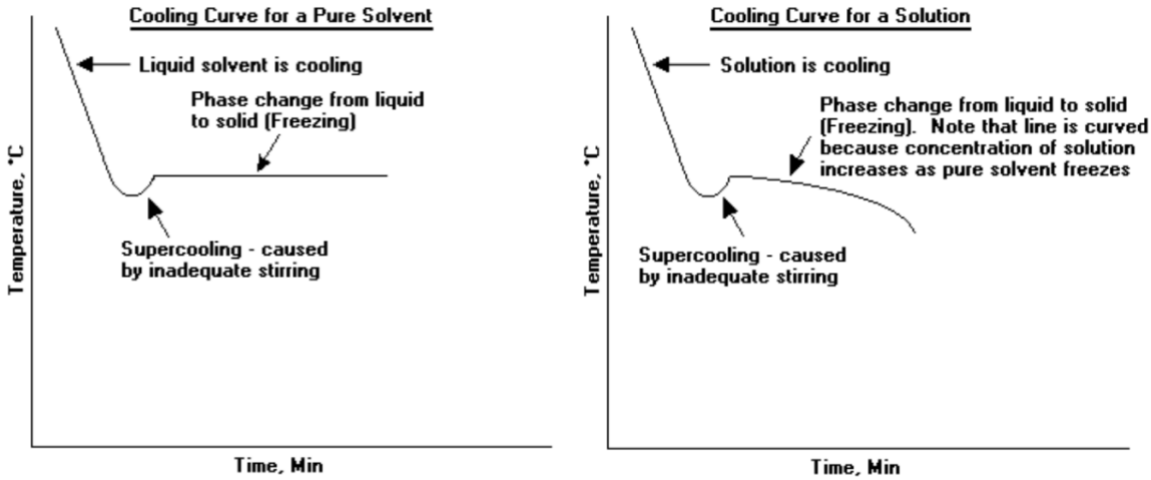


Figure 1. Comparison of cooling curve for a pure solvent (left) to cooling curve for a solution (right).

You will construct cooling curves for both the pure solvent (water) and two solutions.

From the cooling curve you will determine the true freezing temperature. Use Table 1 to record the data collected during the experiment:

Table 2: Recommended table for data collection.

Measured Observations	**Solute: Sucrose**	**Solute: Sodium Chloride**
Temperature of ice-salt bath		
Weight of water (Part D/E)		
Molality of the solution (moles/kg)		
Freezing point of water (Part C)		
Freezing point of solution (Part D/E)		

As the temperature of your solution decreases it will reach a point where supercooling occurs. With continued stirring the temperature will again rise and then slowly decrease as the pure solvent freezes. **The freezing point (T_{fp}) of the solution is the maximum temperature after super cooling has occurred.**

PROCEDURE

A. Calibrating the Temperature Probe

Temperature probes differ somewhat in their electrical responses to temperature, so each probe should be calibrated before using the probe for measurements. IMPORTANT: Use distilled water.

1. Plug in the temperature probe and open Logger Pro.
2. Pull down EXPERIMENT from the Menu panel and select CALIBRATE – GO!LINK: 1 DIRECT CONNECT TEMP.
3. Fill a 250 mL Erlenmeyer about half full of distilled water. Heat the flask until the water is boiling gently. (Assume the temperature of the boiling water to be $94.5\ ℃$ at Durango's elevation.)
4. While this water is heating, prepare a 150 mL beaker of ice and distilled water to make a slurry whose temperature you can assume to be very close to $0.0\ ℃$.
5. Place this beaker on an insulated surface (lab manual, notebook, towel) and let it set for a few minutes to equilibrate.
6. Select CALIBRATE NOW in the Calibration Box and place the probe in either the boiling water or the ice/water slurry and stir the probe around to mix the sample to obtain a representative temperature reading.
7. When the probe reading on the screen shows a steady voltage (V) value, enter the appropriate temperature value and select KEEP.
8. Repeat #7 for the second calibration temperature. Select OK. Then DONE and NO you do not want a third calibration point.
9. You should see the current temperature reading from the probe above the data list.
10. Check your calibration work by reading the temperature of the ice bath and the boiling water bath again, to ensure that your calibration is accurate.
11. If the probe is reading temperatures inaccurately, check your procedure and recalibrate.
12. Adjust the Data Collection parameters with an appropriate sampling rate (maybe 10 samples per minute) to measure changes in temperature in degrees C (℃) versus time in minutes with a maximum time of 30 minutes.

B. Preparation of an Ice Bath

1. Fill a large Styrofoam cup with ice.
2. Add approximately 150 mL of distilled water to the cup.
3. Add approximately 30 g of salt ($NaCl$) to the cup.
4. Stir this ice-salt mixture with a stirring rod until a temperature range of $-6\ ℃$ to $-10\ ℃$ is obtained. NOTE: Colder temperatures are easily obtained by adding more salt, however, it makes the solutions cool too fast to obtain good results.

5. Be sure to check the temperature of the ice bath BEFORE each experiment. You will use this salt-ice bath for the next two parts. If the temperature of the bath rises above -5 ℃, then drain the water and add more ice and salt.

C. Cooling Curve for Pure Solvent Water

1. Obtain a large test tube that has been cleaned and dried. Add approximately 10 mL of distilled water to it.
2. Place the test tube in the salt-ice bath, so that the level of the water is below the level of the ice in the bath.
3. Begin temperature recording by selecting the collect button. Place the temperature probe in the test tube. Continue collecting temperature measurements while vigorously agitating (using the temperature probe with care) the distilled water.
4. Continue to record the temperature until the solution solidifies. Keep collecting data until you obtain 5 – 10 readings after the curve levels out.
5. Save your data to the desktop.
6. From the Menu Bar open ANALYZE choose EXAMINE.
7. Move the line with your mouse to the freezing point. Record the freezing point of water in your data table.
8. Remove the test tube and dump the contents down the drain. You will need to rinse and dry the test tube between trials.
9. Move the line with your mouse to the freezing point. Record the freezing point of water in your data table.
10. Save this cooling curve. The next two you generate can be added on the same plot. These cooling curves should be included it in your data section of your submitted lab notebook.
11. When you start your next data collection, after you hit collect, select Store Latest Run.

D. Determination of (i) for Sucrose

1. Prepare a solution of sucrose ($C_{12}H_{22}O_{11}$) by adding 17 g of sucrose to 50 g of water (weigh both accurately) in a 150 mL beaker. Mix until all crystals have dissolved.
2. Obtain a large test tube that has been cleaned and dried. Add approximately 10 mL of the sucrose solution to it.
3. Place the test tube in the salt-ice bath, so that the level of the sucrose solution is below the level of the ice in the bath.
4. Begin temperature recording by selecting the collect button. Place the temperature probe in the test tube. Continue collecting temperature measurements while vigorously agitating (using the temperature probe with care) the sucrose solution.
5. Continue to record the temperature until the solution solidifies. Note: these solutions tend to form a slush and it is difficult to observe the exact freezing point. Keep collecting data

until you obtain 5 – 10 readings below the freezing point of water. However, if a slush does not form within about 5 min add more salt to the ice bath and ensure the temperature is in the range of $-6℃$ to $-10℃$.

6. Save your data to the desktop.
7. From the Menu Bar open ANALYZE choose EXAMINE.
8. Move the line with your mouse to the freezing point and record the value in your data table.
9. Remove the test tube and dump the contents down the drain. You will need to rinse, and dry the test tube between trials.

E. Determination of (i) for $NaCl$

Repeat Part D using 2.5 g of $NaCl$. Save this cooling curve.

CALCULATIONS

1. Determine ΔT_{fp} for each solution.
2. Use the ΔT_{fp} determined in #1 and Equation (3), see page 2, to calculate the effective i for the sucrose and sodium chloride solutions.

ADDRESS THE FOLLOWING QUESTIONS IN YOUR CONCLUSION

1. Discuss the effects that sucrose and sodium chloride each have on freezing temperature. Are these effects what you expected? Why or why not?
2. Explain what a van't Hoff factor is and why it is important when studying colligative properties.

WASTE AND CLEAN UP

1. All sucrose and salt solutions can be poured down the drain.
2. Clean all glassware and supplies and return them to the supply cart in the same location that you found them.
3. Do not throw away your Styrofoam cups, rinse them and return them to the supply cart.
4. Rewrap the Go!Link and secure it with a rubber band and return it and the adapter to the supply cart.
5. Rewrap the temperature probe and return it to the supply cart.
6. Wipe any mess at your bench top.